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RICE BIOTECHNOLOGY QUARTERLY

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QTLs for seedling vigor

Small inverted repeats discovered

Rice BAC libraries

Microsatellite linked to Waxy

Markers for gall midge resistance

Partial blast resistance

Marker-free vectors developed

Modified toxin gene tested

Water stress and regeneration

Accurate transcription

Microprojectile transformation

PCR-based markers

Seed-specific transcription

Preferential expression of gene



In a recent interview, Dave Mackill (above), USDA-ARS, Department of Agronomy and Range Science, University of California, Davis, stated that the overall objective of their many research projects is to develop improved rice germplasm and solve problems facing the US rice industry through integration of modern genetic technologies with conventional breeding programs. Emphasis is placed on the temperate japonica cultivars adapted to the Western rice growing region (California). The major activities involve germplasm conservation, characterization and utilization, identification of useful genes through linkage to molecular markers, and the development of hybrid rice technology. Genes important for US rice growing environments are identified and tagged with molecular markers. The major target traits include disease (stem rot) and insect (rice water weevil) resistance, cold tolerance, and traits associated with adaptation to water-seeding such as seedling vigor and submergence tolerance. Recent results include use of RAPD markers to classify japonica rice cultivars, identification of quantitative trait loci for seedling vigor traits, development of a RAPD map in a cross between two japonica cultivars, and the tagging of a major submergence tolerance locus in the breeding line IR40931-26. e-mail: djmackill@ucdavis.edu